

# EDAA40 Exam

15 April 2026

## Instructions

### Things you CAN use during the exam.

Any written or printed material is fine. Textbook, other books, the printed slides, handwritten notes, whatever you like.

In any case, it would be good to have a source for the relevant definitions, and also for notation, just in case you don't remember the precise definition of everything we discussed in the course.

### Things you CANNOT use during the exam.

Anything electrical or electronic, any communication device: computers, calculators, mobile phones, toasters, ...

**A sheet with common symbols and notations is attached at the end.**

Good luck!

|    |    |    |    |    |              |
|----|----|----|----|----|--------------|
| 1  | 2  | 3  | 4  | 5  | <b>total</b> |
| 19 | 21 | 16 | 24 | 20 | <b>100</b>   |
|    |    |    |    |    |              |

**Total points: 100**  
**points required for 3: 50**  
**points required for 4: 67**  
**points required for 5: 85**

**1****[19 pt]**

1. [12pt] Let  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  be defined by  $f(x) = x^2$ .

Define a relation  $R$  on  $\mathbb{Z}$  by:

$$xRy \iff f(x) = f(y).$$

- (a) [4pt] Determine which properties  $R$  satisfies (write yes/no below):

Reflexive: \_\_\_\_\_

Symmetric: \_\_\_\_\_

Antisymmetric: \_\_\_\_\_

Transitive: \_\_\_\_\_

- (b) [4pt] Define a function  $g : \mathbb{Z} \rightarrow \mathbb{Z}$  such that  $xRy \iff g(x) = g(y)$  still holds, but  $g$  is not equal to  $f$ .

- (c) [4pt] Consider the claim: “If  $xRy$  is defined as  $g(x) = g(y)$  for some function  $g$ , then  $R$  is always an equivalence relation.” Is the claim correct? Justify.

2. [7pt] Let  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  be defined by  $f(x) = x^2 - x$ .

- (a) [3pt] Is  $f$  injective? Is  $f$  surjective? Is  $f$  bijective? Justify.

- (b) [2pt] Consider  $g : \mathbb{Z} \rightarrow \mathbb{Z}$  defined as  $g(x) = f(x+1) - f(x)$ . Determine the range of  $g$  and justify your answer.

- (c) [2pt] Is  $g$  injective? Surjective? Bijective? Justify.

**2****[21 pt]**

Given sets  $A, B \subseteq \mathbb{Z}$ , let the set  $S_{A,B} \subseteq \mathbb{Z}$  be the set that contains all sums of an element in  $A$  with an element in  $B$ . For example, if  $A = \{0, 1\}$  and  $B = \{6, 10\}$ , then  $S_{A,B} = \{6, 7, 10, 11\}$ .

1. [2pt] Given sets  $A, B \subseteq \mathbb{Z}$ , define the set  $S_{A,B}$  using set builder notation.

$$S_{A,B} :=$$

2. [2pt] Suppose  $A = \{1, 3\}$  and  $B = \{2, 4, 6\}$ . What is the size of  $S_{A,B}$ ? \_\_\_\_\_
3. [9pt] For each item, state whether it is true or false. Justify your answer shortly.

(a)  $\forall A, B \subseteq \mathbb{Z}. (A \neq \emptyset \wedge B \neq \emptyset) \rightarrow \exists x \in S_{A,B}. (\forall a \in A. (x - a \in B))$

(b)  $\exists A \subseteq \mathbb{Z}. (\forall B \subseteq \mathbb{Z}. (S_{A,B} \subseteq B))$

(c)  $\exists A, B \subseteq \mathbb{Z}. (|S_{A,B}| < |A \setminus B|)$

4. [8pt] Let  $A = \{-1, 1\}$  and  $B = \mathbb{N}$ .

(a) [2pt] Describe  $S_{A,B}$  explicitly.

(b) [3pt] Show that  $S_{A,B}$  is countably infinite by defining a bijection between  $S_{A,B}$  and  $\mathbb{N}$ .

(c) [3pt] Show that there exists a bijection between  $S_{A,B}$  and  $\mathbb{Z}$ . You may either define such a bijection explicitly or justify its existence by using a theorem/result about countably infinite sets.

**3****[16 pt]**

Consider the following recursive definition of a function  $f : \mathbb{N} \rightarrow \mathbb{N}$ .

$$f(n) = \begin{cases} 3 & n = 0 \\ 2 & n = 1 \\ f(n-1) + f(n-2) & n \geq 2 \end{cases}$$

You do **not** need to find a closed formula for  $f(n)$  for this exercise.

- (a) [12pt] Prove by (some form of) induction that  $f(n) \leq 3^n$  is true for all  $n \geq 1$ .
- (b) [4pt] Is it possible to prove that  $f(n) \leq 2^n$  is true for all  $n \geq 1$ ? If yes, explain what you would have to change in your proof; if not, explain what step doesn't work.

**4****[24 p]**

Let  $G = (V, E)$  be a directed graph. We say a vertex  $y$  is reachable from a vertex  $x$  if there is a directed path (possibly of length 0) from  $x$  to  $y$ .

## PART I

Your goal is to (recursively) define a function  $f : V \rightarrow \mathcal{P}(V)$  that given a vertex  $v \in V$  outputs the set of all vertices reachable from  $v$ .

Consider the following attempted definition:

$$f(v) := \{v\} \cup \bigcup_{(v,w) \in E} f(w)$$

- (a) [3pt] What is wrong with this attempted definition?
- (b) [10pt] Define a recursive function that avoids this issue.

Hint: define an auxiliary recursive function and then define  $f$  using this auxiliary function. For example, you can define the auxiliary function  $f' : V \times \mathcal{P}(V) \rightarrow \mathcal{P}(V)$  that receives as input a vertex  $v$  and a set  $S$  of already visited vertices, and outputs the set  $S$  union the set of vertices reachable from  $v$  that are not already visited. Then you can define  $f$  in terms of  $f'$ . Or you can use some other auxiliary function — just describe in words your auxiliary function and explain why it is correct.

## PART II

For  $x, y \in V$ , let  $R(x, y)$  be the predicate “ $y$  is reachable from  $x$ .” Express the following three predicate formulas using  $R$  and quantifiers:

- (a) [3pt] There exists a vertex reachable from all vertices.
- (b) [3pt] There exist two vertices that are not mutually reachable (where mutually reachable means that both vertices are reachable from each other).
- (c) [5pt]  $G$  is a directed acyclic graph, that is, there is no directed cycle in  $G$ . Recall that a directed cycle is a directed positive length closed walk whose vertices are distinct except for the beginning and end vertices.

**5****[20 pt]**

Solve either **OPTION A** or **OPTION B** below (not both).

### OPTION A: Number theory

1. [8pt] Why is a number divisible by 3 if and only if the sum of its digits (when written in decimal) is a multiple of 3?

2. [12pt] In class we proved the following theorem.

**Theorem 1.** *The greatest common divisor of  $a$  and  $b$  can be written as a linear combination of  $a$  and  $b$ . That is,*

$$\gcd(a, b) = sa + tb$$

*for some integers  $s$  and  $t$ . Moreover,  $\gcd(a, b)$  is the smallest positive integer that can be written as a linear combination of  $a$  and  $b$ .*

For positive integers  $a, b, c$ , state whether the following statements are TRUE or FALSE and then prove or give a counter-example.

You may not appeal to uniqueness of prime factorization (because some of these properties are needed to prove unique factorization) but you can use the theorem above.

- (a) If  $\gcd(a, b) = 1$  and  $\gcd(a, c) = 1$ , then  $\gcd(a, b^2c) = 1$
- (b) If  $a|bc$  and  $b|c$ , then  $a|c$
- (c) If  $a|b$  and  $b|c$ , then  $\gcd(a, c) = a$
- (d) If  $\gcd(a, b) = 1$ , then  $a + b$  and  $ab$  are coprime

## OPTION B

1. [14pt] When working with sets, we are sometimes interested in considering subsets of some set containing all the elements we care about, that is, all the elements in some *local universe* which we denote by  $U$ .

- (a) [2pt] Given  $A, B \subseteq U$ , define the symmetric difference of  $A$  and  $B$  (all elements that are in  $A$  or  $B$  but not in both) using set builder notation. You can use negation  $\neg$ , OR  $\vee$ , AND  $\wedge$ , and “element of”  $\in$  (and “not an element of”  $\notin$ ).

$$A \triangle B :=$$

- (b) [8pt] It is known that all set operations can be expressed using only set intersection and set difference. Express each of the three operations (complement, union and symmetric difference) using only intersection ( $\cap$ ) and difference ( $\setminus$ ). You can use the local universal set  $U$ .

$$\overline{A} =$$

$$A \cup B =$$

$$A \triangle B =$$

- (c) [4pt] It is known that all set operations defined above can be expressed using only set difference. Express intersection using only difference ( $\setminus$ ). (Hint: you’ll need to use the set  $U$ .)

$$A \cap B =$$

2. [6pt] Write the following two logical formulas in disjunctive normal form (DNF). Explain how you obtained the DNF formula.

(a)  $p \leftrightarrow (q \wedge r)$

(b)  $(p \vee q) \rightarrow \neg(p \wedge \neg r)$

## Some common symbols

- $\mathbb{N}$  the natural numbers, starting at 0
- $\mathbb{N}^+$  the natural numbers, starting at 1
- $\mathbb{Z}$  the integers
- $\mathbb{R}$  the real numbers
- $\mathbb{Q}$  the rational numbers
- $a \perp b$   $a$  and  $b$  are coprime, i.e., they do not have a common divisor other than 1
- $a \mid b$   $a$  divides  $b$ , i.e.,  $\exists k \in \mathbb{Z}$  such that  $ka = b$
- $\mathcal{P}(A)$  power set of  $A$
- $\overline{R}$  of a relation  $R$ : its complement
- $R^{-1}$  of a relation  $R$ : its inverse
- $R \circ S, f \circ g$  of relations and functions: their composition or a function  $f$
- $[a, b], ]a, b[, ]a, b], [a, b[$  closed, open, and half-open intervals from  $a$  to  $b$
- $A \sim B$  two sets  $A$  and  $B$  are equinumerous
- $A^*$  for a finite set  $A$ , the set of all finite sequences of elements of  $A$ , including the empty sequence,  $\varepsilon$
- $\sum S$  sum of all elements of  $S$
- $\prod S$  product of all elements of  $S$
- $\bigcup S$  union of all elements of  $S$
- $\bigcap S$  intersection of all elements of  $S$
- $\bigcup_{a \in S} E(a), \bigcap_{a \in S} E(a)$  generalized union / intersection of the sets computed for every  $a$  in  $S$